Giovanni Tauro

R&D Engineer| Sensor Data, Python & Data Quality



Bari-Lecco, Italy +39 3279437161 giovanni.taurodi@gmail.com
LinkedIn GitHub Scholar Portfolio GrabCAD

R&D Engineer and PhD Candidate with 5+ years at CNR-STIIMA and Head of Product of Rehabilia Technologies. For OmniTrack I own the data path end to end: UWB and IMU time series from wearable tags, session logging in MongoDB, and Python analysis (pandas, Jupyter) that turns study recordings into validated datasets and trial metrics. I care about data correctness: raw records are never modified and every correction is traceable. I use Claude and Claude Code daily to move faster in analysis and prototyping.

Work Experience

R&D Product Engineer, STIIMA, Lecco
Feb 2021 – Present

- Built wearable and IoT sensing systems for rehabilitation research, collecting time-series data (position, IMU) in studies with patients and clinicians.
- Defined the architecture of multimodal IoT platforms, integrating wearable sensing, indoor localization, BLE communication, and data logging.
- Designed evaluation protocols and analysed usability, system performance, and user interaction data in Python.
- Defined hardware/software requirements with clinicians and deployed PhiCube devices in Italian clinics within Fit4MedRob (MUR, national programme).

`Python • pandas • NumPy • MongoDB • ROS2 • ESP32 • BLE GATT • C++ • Git • Linux`

Head of Product (Co-Founder), Rehabilia Technologies s.r.l., Milan
May 2023 – Present

- Lead product development and prioritisation with engineering, design and clinical stakeholders, from user research to validation.
- Run ongoing research with clinics and education centres on their needs, the tools they use, and how they make spending and purchasing decisions, to inform product and business planning.
- Validate decisions with study data: PhiCube reached SUS 85.8/100 with 19 children and is in two ongoing multicentric studies with 134 patients.

`Prioritisation • Customer Research • Study Data • Roadmaps`

Selected Projects



GIT_1
GIT_2
GIT_3

OmniTrack: IoT-based Multimodal Navigation System /Frontiers

- Built the data path end to end: UWB ranging and IMU time series from wearable ESP32 tags, position estimation in ROS 2, and session logging in MongoDB.
- Designed OmniAnalytics, a data tool that turns raw session recordings from 16 blind participants into trial metrics: route similarity (DTW, Fréchet distance), speed and target discovery.
- Kept data traceable: corrections, exclusions and annotations are stored separately, so raw recordings are never modified.

`Python` • `pandas` • `Jupyter` • `MongoDB` • `ROS2` • `ESP32` • `UWB` • `IMU`



WEB

PhiCube: reconfigurable bilateral robotic device for neurorehabilitation /Frontiers

- Took a modular rehabilitation device from user research and prototypes to clinical use.
- Validated with 19 children (SUS 85.8/100); 134 patients in two on-going multicentric studies.

`Prototyping` • `Usability` • `Clinical Studies` • `Patent`

Education

2023–27 **PhD, Mechanics and Advanced Engineering Sciences** *Bologna University, IT*

PhD Candidate, expected 2027. Topic: IoT-based Multimodal Navigation System – OmniTrack

- Research Mobility, Utrecht University, NLD: Sensory navigation and experimental psychology.
- Research Mobility, Toyo University, JP: Feedback algorithm and behaviour analysis.

2017–20 **MSc, Design & Engineering** *Politecnico di Milano, IT*

Topic: UX development of an ultrasonic dental device: from concept to validation: WashB

- Research Mobility, Chiba University, JP: UX research studies for Fujitsu and Panasonic.

2012–15 **BSc, Industrial Design** *Florence University, IT*

Topics: Foundation in research methodologies, development, and critical thinking.

Skills

Data Python (pandas, NumPy, Jupyter), MongoDB, time-series processing, data validation

Analysis Metric definition, evaluation protocols, study data analysis, customer research

Tools Git, Linux, C++, ROS 2, Claude, Claude Code

Courses & Certifications

2026 **Claude 101 Badge — Claude Code Badge — Claude** *Anthropic, Online*

2023 **Sketching with Hardware: Prototyping Tangible AI** *LMU Munich, DE*

2022 **Experimental Interaction Design** *Tallinn University, EE*

Languages

Italian: Native English: Advanced (IELTS 2017) French: Basic (A2)

Lecco, October 18, 2026



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